

THE PHYSICAL AI CO-DESIGN MATRIX

Twenty-five questions. Five cognitive obligations against five physical constraints. Part II answers them one row at a time.

	C1 Time & Freshness	C2 Inertia & Stopping	C3 Actuation Limits	C4 Energy & Thermal	C5 Silicon Determinism
PERCEIVE Ch 4 · OBS-01	When did this observation happen?	How far ahead must you see?	<i>not binding</i> commands nothing	What does sensor heat do to geometry?	DOMINANT What does moving the pixels cost?
REMEMBER Ch 5 · STATE-01	DOMINANT What is a belief worth after Δt ?	How much world must you hold?	<i>thin</i> relocalization snap	Which calibration drifts with heat?	Bounded state without allocation
REASON Ch 6 · INTENT-01	DOMINANT How long may intent outlive its world?	<i>not binding</i> reaches via the lease	<i>not binding</i> commands no torque	What does throttling do to a deadline?	Keeping a large model off the real-time core
PLAN Ch 7 · PLAN-01	Outliving the latency that produced it	What makes a plan abandonable?	DOMINANT What shapes may a command take?	What does a motion cost per shift?	<i>thin</i> compute as a budget
EXECUTE Ch 8 · ENF-01	The deadline, and the cycle it is missed	DOMINANT How much margin does stopping need?	Can the veto violate the limit it protects?	What clamps the current asked for?	Code that may never pause

READING THE GRID: **Bold border** marks each chapter's dominant cell, where most of its length goes. **Shaded** cells do not bind, and saying why is part of the argument. The grid is uneven because the physics is.